



gemma4:e2b

↓ 13.7M Downloads ⌚ Updated 3 days ago

Gemma 4 models are designed to deliver frontier-level performance at each size. They are well-suited for reasoning, agentic workflows, coding, and multimodal understanding.

- vision
- tools
- thinking
- audio
- cloud
- e2b
- e4b
- 12b
- 26b
- 31b

CLI

cURL

Python

JavaScript

ollama run gemma4:e2b

Details

Updated 2 months ago			7fbdbf8f5e45 · 7.2GB ·
model	arch gemma4 · parameters 5.12B · quantization Q4_K_M	7.2GB	
license	Apache License Version 2.0, January 2004 http://www.ap...	11kB	
params	{ "temperature": 1, "top_k": 64, "top_p": 0.95 }	42B	

Readme



Gemma 4

Gemma is a family of open models built by Google DeepMind. Gemma 4 models are multimodal, handling text and image input and generating text output.

Gemma 4 introduces key **capability and architectural advancements**:

- **Reasoning** – All models in the family are designed as highly capable reasoners, with configurable thinking modes.
- **Extended Multimodalities** – Processes Text, Image with variable aspect ratio and resolution support (all models)
- **Diverse & Efficient Architectures** – Offers Dense and Mixture-of-Experts (MoE) variants of different sizes for scalable deployment.
- **Optimized for On-Device** – Smaller models are specifically designed for efficient local execution on laptops and mobile devices.
- **Increased Context Window** – The small models feature a 128K context window, while the medium models support 256K.
- **Enhanced Coding & Agentic Capabilities** – Achieves notable improvements in coding benchmarks alongside native function-calling support, powering highly capable autonomous agents.
- **Native System Prompt Support** – Gemma 4 introduces native support for the `system` role, enabling more structured and controllable conversations.

Models

Ollama's cloud

```
ollama run gemma4:31b-cloud
```

Edge models

The "E" in E2B and E4B stands for "effective" parameters, and are made for edge device deployments.

Effective 2B (E2B)

```
ollama run gemma4:e2b
```

Effective 4B (E4B)

```
ollama run gemma4:e4b
```

Workstation models

These models are designed for frontier intelligence locally.

12B

```
ollama run gemma4:12b
```

26B (Mixture of Experts model with 4B active parameters)

```
ollama run gemma4:26b
```

31B (Dense)

```
ollama run gemma4:31b
```

Benchmark Results

These models were evaluated against a large collection of different datasets and metrics to cover different aspects of text generation. Evaluation results marked in the table are for instruction-tuned models.

	Gemma 4 31B	Gemma 4 26B A4B	Gemma 4 E4B	Gemma 4 E2B	Gemma 3 27B (no think)
MMLU Pro	85.2%	82.6%	69.4%	60.0%	67.6%
AIME 2026 no tools	89.2%	88.3%	42.5%	37.5%	20.8%
LiveCodeBench v6	80.0%	77.1%	52.0%	44.0%	29.1%
Codeforces ELO	2150	1718	940	633	110
GPQA Diamond	84.3%	82.3%	58.6%	43.4%	42.4%
Tau2 (average over 3)	76.9%	68.2%	42.2%	24.5%	16.2%
HLE no tools	19.5%	8.7%	-	-	-
HLE with search	26.5%	17.2%	-	-	-
BigBench Extra Hard	74.4%	64.8%	33.1%	21.9%	19.3%
MMMLU	88.4%	86.3%	76.6%	67.4%	70.7%
Vision					
MMMU Pro	76.9%	73.8%	52.6%	44.2%	49.7%
OmniDocBench 1.5 (average edit distance, lower is better)	0.131	0.149	0.181	0.290	0.365
MATH-Vision	85.6%	82.4%	59.5%	52.4%	46.0%
MedXPertQA MM	61.3%	58.1%	28.7%	23.5%	-
Audio					
CoVoST	-	-	35.54	33.47	-
FLEURS (lower is better)	-	-	0.08	0.09	-
Long Context					

	Gemma 4 31B	Gemma 4 26B A4B	Gemma 4 E4B	Gemma 4 E2B	Gemma 3 27B (no think)
MRCR v2 8 needle 128k (average)	66.4%	44.1%	25.4%	19.1%	13.5%

Model information

Property	E2B	E4B	31B Dense
Total Parameters	2.3B effective (5.1B with embeddings)	4.5B effective (8B with embeddings)	30.7B
Layers	35	42	60
Sliding Window	512 tokens	512 tokens	1024 tokens
Context Length	128K tokens	128K tokens	256K tokens
Vocabulary Size	262K	262K	262K
Supported Modalities	Text, Image, Audio	Text, Image, Audio	Text, Image
Vision Encoder Parameters	~150M	~150M	~550M
Audio Encoder Parameters	~300M	~300M	No Audio

Mixture-of-Experts (MoE) Model

Property	26B A4B MoE
Total Parameters	25.2B
Active Parameters	3.8B
Layers	30
Sliding Window	1024 tokens

Property	26B A4B MoE
Context Length	256K tokens
Vocabulary Size	262K
Expert Count	8 active / 128 total and 1 shared
Supported Modalities	Text, Image
Vision Encoder Parameters	~550M

Best Practices

For the best performance, use these configurations and best practices:

1. Sampling Parameters

Use the following standardized sampling configuration across all use cases:

- temperature=1.0
- top_p=0.95
- top_k=64

2. Thinking Mode Configuration

Note that Ollama already handles the complexities of the chat template for you.

Compared to Gemma 3, the models use standard system , assistant , and user roles. To properly manage the thinking process, use the following control tokens:

- Trigger Thinking:** Thinking is enabled by including the <|think|> token at the start of the system prompt. To disable thinking, remove the token.
- Standard Generation:** When thinking is enabled, the model will output its internal reasoning followed by the final answer using this structure:
<|channel>thought\n [Internal reasoning] <channel|>
- Disabled Thinking Behavior:** For all models except for the E2B and E4B variants, if thinking is disabled, the model will still generate the tags but with an empty thought block:
<|channel>thought\n<channel|> [Final answer]

3. Multi-Turn Conversations

- **No Thinking Content in History:** In multi-turn conversations, the historical model output should only include the final response. Thoughts from previous model turns must *not be added* before the next user turn begins.

4. Modality order

- For optimal performance with multimodal inputs, place image and/or audio content **before** the text in your prompt.

5. Variable Image Resolution

Aside from variable aspect ratios, Gemma 4 supports variable image resolution through a configurable visual token budget, which controls how many tokens are used to represent an image. A higher token budget preserves more visual detail

at the cost of additional compute, while a lower budget enables faster inference for tasks that don't require fine-grained understanding.

- The supported token budgets are: **70, 140, 280, 560, and 1120**.
 - Use *lower budgets* for classification, captioning, or video understanding, where faster inference and processing many frames outweigh fine-grained detail.
 - Use *higher budgets* for tasks like OCR, document parsing, or reading small text.